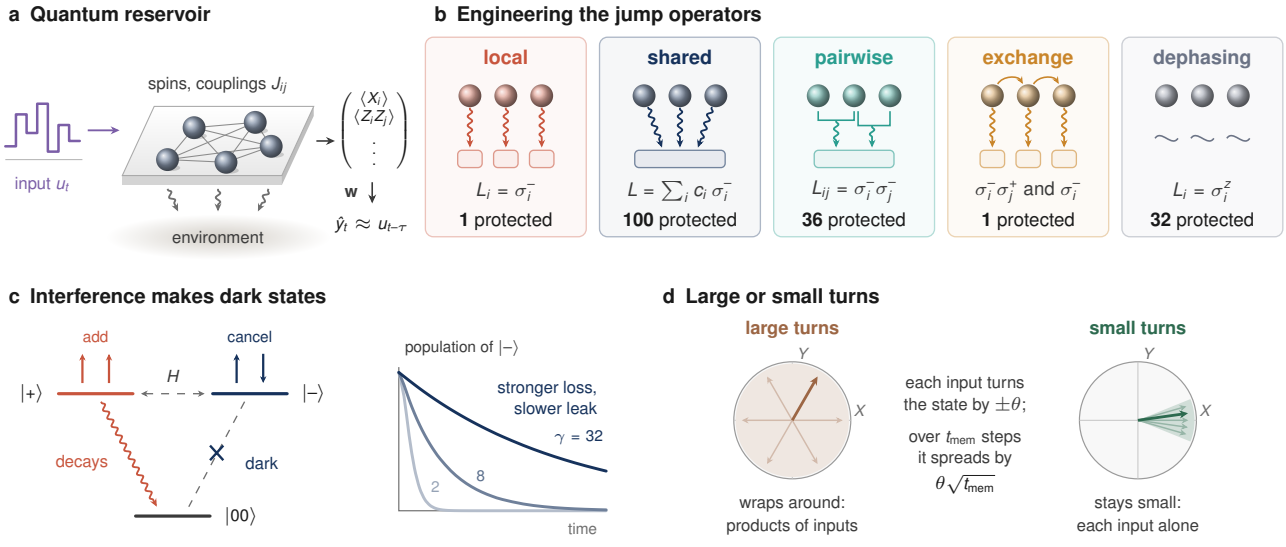

a Quantum reservoir
spins, couplings J_{ij}
input u_t
environment
$\langle X_i \rangle$
$\langle Z_i Z_j \rangle$
w
$\hat{y}_t \approx u_{t-\tau}$
b Engineering the jump operators
local
$L_i = \sigma_i^-$
1 protected
shared
$L = \sum_i c_i \sigma_i^-$
100 protected
pairwise
$L_{ij} = \sigma_i^- \sigma_j^-$
36 protected
exchange
$\sigma_i^- \sigma_j^+$ and σ_i^-
1 protected
dephasing
$L_i = \sigma_i^z$
32 protected
c Interference makes dark states
add
cancel
H
$|+\rangle$
$|-\rangle$
decays
dark
$|00\rangle$
population of $|-\rangle$
stronger loss,
slower leak
$\gamma = 32$
8
2
time
d Large or small turns
large turns
Y
X
wraps around:
products of inputs
each input turns
the state by $\pm\theta$;
over t_{mem} steps
it spreads by
$\theta\sqrt{t_{\text{mem}}}$
small turns
Y
X
stays small:
each input alone